MIL-P-13988A(MU)
2 July 1970
SUPERSEDING
MIL-P-13988(Ord)
7 March 1955

MILITARY SPECIFICATION

PAPER, LENS, TISSUE, ANTITARNISH, WRAPPING

1. SCOPE

2. This specification covers one type of tissue paper, of antitarnish quality, for use in wrapping lenses.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Steel Strapping, Flat
UU-P-31 - Paper, General Specifications and Methods of Testing
UU-P-268 - Paper, Kraft, Untreated, Wrapping
PPP-B-601 - Box, Wood, Cleated-Plywood
PPP-B-636 - Box, Fiberboard
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

FSC 6640

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:44Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies)

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY

TAPPI Method T444m-47 - Silver-Tarnishing Test of Paper

(Copies of the above cited method may be obtained from the Technical Association of the Pulp and Paper Industry, 360 Lexington Ave., New York, N. Y. 10017).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, D. C. 20036.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:44Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material. The tissue paper shall be made from either white or unbleached wood pulp, free of ground wood particles.

3.2 Physical and chemical properties. The tissue paper shall conform to the physical and chemical requirements specified hereinunder, when tested in accordance with the test methods specified in section 4 of this specification.

3.2.1 Basis weight. The basis weight of the tissue paper (20 by 30-500) shall be 8.0 ± 0.5 pounds.

3.2.2 Bursting strength. The average bursting strength of the tissue paper shall be not less than 5 pounds.

3.2.3 Acidity or alkalinity (pH value). The pH value of the tissue paper shall be in the range of 6.0 to 7.0. The tissue paper shall contain no free acids or alkali and shall be free of chlorides and sulfur.

3.2.4 Tarnish resistance. The tissue paper shall not cause tarnishing of silver surfaces.

3.2.5 When tested as specified in 4.4.3, the tissue paper shall not cause any scratches to the surface of polymethyl methacrylate.

3.3 Workmanship. The tissue paper shall be free of abrasive materials, lint, and dust. The edges of the sheets shall be clean-cut.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:44Z
Check the source to verify that this is the current version before use.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of all tissue paper produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous.

4.2.2 Sampling. Sampling shall be in accordance with the applicable provisions of method 160 of UU-P-31. The number of test specimens taken from the lot shall be required by the individual tests to which the tissue paper is to be subjected (see 4.4).

4.3 Inspection.

4.4 Tests.

4.4.1 The following tests shall be performed in accordance with the indicated test methods of UU-P-31. The tests shall be performed under standard testing atmospheric conditions, as specified in method 102 of UU-P-31.

Test	Method No.
Fiber identification	300
Basis weight	110
Bursting strength[1]	112
Acidity, pH	200

[1]The average bursting strength of 10 test specimens shall be reported.

4.4.2 Tarnish resistance. The tarnish resistance quality of the tissue paper shall be determined in accordance with method T444m-47 of the Technical Association of Pulp and Paper Industry.

4.4.3 Test for abrasiveness. The apparatus shall consist of a methacrylate plate, and a rotating roller having a metal surface such that it will hold a strip of paper in close contact with the plate under a pressure of 1 kilogram per centimeter width of the paper. The plate and roller shall be clean and free of dust, and the plate shall have no scratches on its surface. A strip of the paper to be tested shall be drawn slowly through a distance of 10 cm. between the plate and the roller. The plate shall be examined for scratches under a microscope, illuminated from below at an angle of 95°.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:44Z
Check the source to verify that this is the current version before use.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completely processed pack prepared for shipment shall be considered a unit or product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 defects per hundred units.

(a) Materials, methods and containers not as specified. Each incorrect material, method or container shall be considered one defect.
(b) Intermediate packaging not as specified.
(c) Intermediate container not waterproofed as specified.
(d) Strapping not zinc coated for level A.
(e) Marking illegible, incorrect or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. One ream (480 sheets) of tissue paper shall be wrapped in kraft paper conforming to UU-P-268 and sealed with tape conforming to PPP-T-45. At option of the contractor, a shrink-pack made with polyvinyl chloride, cellulose acetate or cellulose butyrate may be used.

5.1.1.1 Intermediate package. Tissue paper, packaged as specified, shall be intermediate packaged, as follows: full size sheets, 5 reams; half-size sheets, 10 reams; and quarter-size sheets, 20 reams; flat in a snug-fitting fiberboard container conforming to PPP-B-636, V3c or V3s. Each container shall be closed, and waterproofed by means of tape in accordance with the appendix of the container specification.

5.1.2 Level C. The tissue paper shall be packaged in standard commercial quantities in a manner to afford protection against deterioration and damage from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:44Z
Check the source to verify that this is the current version before use.

5.2.1 Level A. Tissue paper packaged as specified in 5.1 shall be packed in containers conforming to PPP-B-601, overseas type, style I, in quantities not to exceed the weight limitations of the container. Strapping shall be zinc coated.

5.2.2 Level B. Tissue paper, intermediate packaged as specified in 5.1.1.1 shall require no additional packing. The intermediate package shall be strapped with 2 girthwise straps conforming to QQ-S-781, type I or IV, class B, 3/8 by 0.015 minimum. Tissue paper packaged as specified in 5.1.2 shall be packed in a container as specified in 5.1.1.1 and strapped as above.

5.2.3 Level C. Tissue paper, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery to destination at lowest rates, in compliance with UFC rules or NMFC rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military requirements. In addition to any special marking by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tissue paper is to be used for wrapping lenses and other glass items and highly polished surfaces.

6.2 Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing. (see 5.1 and 5.2).

6.3 Sheet sizes. Two basic sheet sizes produced by the industry are 20 by 30 inches and 24 by 36 inches. Multiple, half and quarter sizes of the basic sizes may also be procured.

Custodian:
Army - MU

Preparing activity:
Army - MU

Review activities:
Army - GL, MU, ME, WC

Project No. 6640-A636

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:44Z
Check the source to verify that this is the current version before use.